TEXORA · LECTURE SHEET

Plain Weave and Its Derivatives

Plain weave interlaces every end with every pick — the firmest, simplest and most produced structure in the world. Its derivatives keep the plain interlacing but group the threads: warp rib, weft rib and hopsack (matt) weaves, which trade a little firmness for bold ribbed and chequered surfaces.

TE-3205 · FABRIC STRUCTURE AND DESIGN · TOPIC 2 · LEVEL 3 TERM 2

1. Plain weave (1/1)

Each end passes alternately over and under successive picks, and adjacent ends work in opposition: end 1 is up on pick 1, end 2 is down, end 3 up, and so on (Figure 1 of Topic 1). The repeat is **2 ends × 2 picks**, the smallest of any weave, and the interlacing frequency is the maximum possible — every thread changes side at every crossing. Consequences, all of which return as examination points:

- **Firmness and stability:** the cloth is firm, does not slip easily at seams, and keeps its structure in finishing and wear better than any float weave of the same yarns and sett.
- **Equal appearance both sides** (with similar warp/weft counts and colours) — there is no distinct face weave effect; the face is decided by density and yarn quality instead.
- **Maximum crimp:** both systems bend at every crossing, so plain weave gives the highest crimp (commonly of the order of 8–12% in cotton cloth) and correspondingly the greatest weaving shrinkage and finishing contraction. It also limits the maximum sett: threads jam against each other sooner than in float weaves.
- **Production:** two shafts suffice in principle (Topic 1, Figure 2); in practice 4 shafts are common at high end densities so that each shaft carries fewer ends and friction stays low — the draft is straight, ends distributed in order over the shafts, and the peg plan follows the plain order on shaft pairs.

Uses are endless: shirting poplin and cambric, sheeting, dress fabrics, organdie, industrial base cloths. Where the examination asks for a comparison, plain weave is always the reference: most interlacing, most crimp, firmest cloth, least lustre (floats are of length 1, so no light is reflected in organised bands).

2. The derivative idea: make threads work in groups

All plain-weave derivatives keep the rule "up, down, up, down" but apply it to **groups** of ends or picks acting as one unit. Grouping warp ends gives ribs running along the fabric (weft-way cords); grouping

picks gives ribs running across it (warp-way cords); grouping both gives chequer blocks. The interlacing is still plain — only the unit size changes — so these fabrics remain firm and stable, with a bold surface texture from the grouping.

3. Warp rib weave

Two (or more) adjacent ends lift together over one pick and go down together for the next: the weft is covered by pairs of ends on one face for one pick, then the pairs exchange. The repeat of the regular **2/2 warp rib** is **4 ends × 2 picks**.

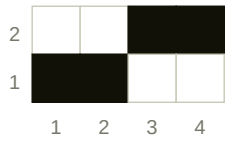


Figure 1 — Warp rib 2/2 (one repeat: 4 ends × 2 picks). Ends 1–2 rise and fall as one unit, ends 3–4 as the other; the weft, hidden under pairs of ends in turn, throws fine cords across the fabric (weft-way ribs).

Because pairs of ends cover the weft, the weft count and quality can be lower than the warp's without showing: warp rib is a **warp-faced derivative**. **Irregular warp rib** mixes group sizes (e.g. groups of 2 and 3 ends alternating: repeat 10 ends × 2 picks), giving uneven cord widths. Typical fabrics: grosgrain and ottoman (heavier cords, silk or filament warps), poplin-type shirtings in fine form. Denting normally puts each group in one dent, emphasising the cord.

4. Weft rib weave

The mirror construction: picks act in pairs. Two successive picks pass over the same ends and under the same ends, the pairs alternating: repeat of the regular **2/2 weft rib** is **2 ends × 4 picks**.

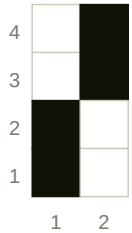


Figure 2 — Weft rib 2/2 (one repeat: 2 ends × 4 picks). Picks 1–2 interlace identically, as do picks 3–4 in opposition; the warp is covered in pairs of picks, producing warp-way cords and a weft-faced surface.

Weft rib is **weft-faced**: the weft dominates the surface, so a cheaper or softer weft can form the visible face over a stronger warp. Irregular weft ribs vary the pick-group sizes. The classic example is **faille** and heavier dress fabrics; in cotton, weft-rib constructions appear in furnishing cloths.

5. Hopsack (matt / basket) weave

Both systems group: in **2/2 hopsack**, two ends and two picks act as units in plain order, giving a chequerboard of 2 × 2 blocks on a repeat of **4 × 4**.

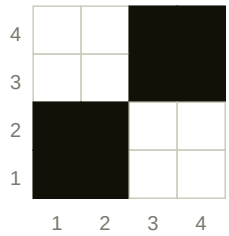


Figure 3 — Hopsack (matt) 2/2, one repeat 4 × 4. Each 2 × 2 block is warp-up or weft-up as a whole. The design is produced either with grouped ends on 2 shafts or with ends drawn singly on 4 shafts lifted in pairs; denting is 2 ends per dent to keep the pairs together.

The grouped threads lie side by side and flatten, so hopsack fabrics are **more open and more extensible** than plain weave of the same sett, with a characteristic small chequer texture; they are also more liable to seam slippage because each unit can shift as a block. Uses: suiting and dress fabrics (often wool or blended), monk's cloth, and — in coarse cotton with heavy yarns — the traditional **basket weave** of furnishings, which is the same interlacing drawn and dented as visible pairs. Larger units (3/3, 4/4 hopsack) increase openness and slippage further and need correspondingly firmer yarns and finishing. The naming point to keep: *hopsack, matt and basket weave are one interlacing*; the trade names differ by yarn, sett and finishing, not by design.

6. The family compared

STRUCTURE	REPEAT	UNIT GROUPED	SURFACE	TYPICAL FABRICS
Plain 1/1	2 × 2	—	Balanced, matt	Poplin, cambric, sheeting
Warp rib 2/2	4 × 2	Ends in 2s	Weft-way cords, warp-faced	Grosgrain, ottoman
Weft rib 2/2	2 × 4	Picks in 2s	Warp-way cords, weft-faced	Faille, furnishing cloth
Hopsack 2/2	4 × 4	Ends and picks in 2s	Chequer blocks, opener handle	Suitings, monk's cloth, basket weave

WORKED CHECK — REPEATS OF THE RIB FAMILY

Regular warp rib with groups of 3 ends: ends repeat in two groups = 6 ends; picks still alternate singly in effect, repeat 2 → **6 × 2**. Irregular warp rib with groups of 2 and 3 alternating: one full pattern of groups is 2 + 3 = 5 ends, and the grouping must repeat twice before the design tiles against its own start (the second group-pair begins with the opposite pick) → **10 ends × 2 picks**. The rule: the repeat must contain the grouping pattern an even number of times when the base unit alternates every pick.

Glossary

Warp rib — plain derivative with ends grouped; weft-way cords. **Weft rib** — picks grouped; warp-way cords. **Regular / irregular rib** — equal / unequal group sizes. **Hopsack (matt)** — both systems grouped in plain order; chequer surface. **Warp-faced / weft-faced** — the face dominated by warp / weft cover.

Key points

• Plain weave: repeat 2 × 2, maximum interlacing, maximum crimp, firmest and most stable structure.
• Derivatives group the unit, not the rule: warp rib 4 × 2, weft rib 2 × 4, hopsack 4 × 4.
• Grouping trades a little stability for texture: cords (ribs) or chequers (hopsack); slippage rises as the unit grows.

Practice

1. Draw (on squared paper, in your notebook) the design for irregular warp rib with end groups 3, 2, 3, 2 and state its repeat.
2. A fabric must show fine cords across its width, with the visible face formed almost entirely by the warp. Choose the structure and justify the choice in two lines.
3. Why does hopsack 3/3 slip at the seams more readily than plain weave woven from the same yarns at the same sett?
4. State the peg plan, on a straight draft over 4 shafts with ends lifted in pairs (1,2) and (3,4), for hopsack 2/2.

Answer key

1. Groups (3+2) repeated twice = 10 ends; picks 2. Design: picks alternate — on pick 1 the first 3 ends up, next 2 down, next 3 up, last 2 down; pick 2 the exact reverse. Repeat **10 × 2**.
2. Warp rib: ends grouped in pairs (or more) rise and fall together, covering the weft on the face for each pick in turn, so the warp forms both the cords (which run weft-way, across the width) and the

face.

3. In hopsack each interlacing unit is a block of 3 parallel threads: the block can slide as a whole against its neighbour because only the block edges interlace; threads per interlacing point are three times fewer than in plain weave at the same sett, so the grip per thread is much lower.
4. Picks 1–2: shafts 1 and 2 lifted (ends 1–2 up), shafts 3–4 down. Picks 3–4: shafts 3 and 4 lifted, 1 and 2 down. The plan is a 2 × 2 block pattern over the four shafts — the peg plan has the same block form as the design.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib